

Physics-Informed ML for Magnetic Reconstruction from XMCD Scattering

Results Summary — ECE 228 Final Project

1. Simulation & Beamline Parameters

Parameter	Value	Parameter	Value
Photon energy	707.0 eV	Detector size	512 x 512 px
Wavelength	1.7537 nm	Pixel size	48.0 um
Wavevector k	3.5829 rad/nm	Sample-detector dist.	150.0 mm
Theta range	15-60 deg (8 steps)	Phi positions	0, 45, 90, 135 deg
Total angle pairs	32	Volume grid	140 x 140 x 3
Absorption length	30.0 nm	XMCD asymmetry	0.30
Poisson noise	Off	Sample scale	75 nm/unit (~1.5 um FOV)

2. Reciprocal Space Coverage

Axis	Range (rad/nm)	Notes
q_x	[-0.2924, 0.2763]	Lateral
q_y	[-0.2924, 0.2763]	Lateral
q_z	[2.5344, 7.0067]	Depth sensitivity; delta_qz = 4.47 rad/nm -> depth res ~1.40 nm
Missing wedge	theta < 15 deg	Limits depth resolution at low q_z

3. Sensing Matrix Diagnostics (Module 3)

Metric	Value	Metric	Value
Dimensions	32 ch x 3 components	Rank	3 / 3 (full)
Condition number	3.62	cond(P ^A T P)	13.13
Singular values	4.897, 2.489, 1.352	Relative SVs	1.000, 0.508, 0.276
Mx visibility	0.411	My visibility	0.467
Mz visibility	0.571	Status	OK — well-conditioned

4. RAAR Phase Retrieval Seed (Module 4)

Setting / Metric	Value	Setting / Metric	Value
Iterations/channel	300	Beta	0.870
Shrink-wrap	Every 25 iters, s=2.0, t=0.15	Runtime	86.3 s
Mean LS residual	0.9511	Channels	32
L2 error (seed)	100.00 %	Mean angular error	90.93 deg
Median angular error	90.27 deg	Assessment	Effectively random

5. Direct Least-Squares Seed (module4_ls.m)

Metric	Value	Metric	Value
Method	LS from b_minus amplitudes	Depth layers	z = 0, 225, 450 nm
Depth weights	0.60 (uniform)	Random baseline	~90 deg
L2 error (seed)	95.48 %	Mean angular error	54.97 deg
Median angular error	42.32 deg	Assessment	Better than random; usable seed

6. RESIRE-V Refinement (Module 6)

Setting / Metric	Value	Setting / Metric	Value
Optimizer	Adam	LR / schedule	1e-5 / cosine
Beta1 / Beta2	0.90 / 0.999	Iterations	300
Support	Off	Regularization	0.0
Self-check residual	0.342 (FAIL)	Runtime	418.4 s
Amplitude err (it=1)	1.8191e+03	Amplitude err (it=300)	1.7803e+03

7. Final Reconstruction Results — Is_direct seed + RESIRE-V Adam

Metric	Global	z=1 (0 nm)	z=2 (3 nm)	z=3 (6 nm)
Relative L2 error (%)	83.50	72.43	95.39	81.06
Mean angular error (deg)	47.76	42.11	54.94	46.24
Median angular error (deg)	41.40	41.14	42.22	41.22

Component	RMSE	Percent Error	Masked pixels
Mx	0.5357	35.19 %	29,400
My	0.5643	44.27 %	29,400
Mz	0.3031	57.69 %	58,520
Amplitude residual	1.7803e+03	—	—

8. ML Architecture Benchmark (MagXMCD_v9 — Angle-Conditioned, 30 epochs)

Model	MSE Total	MSE Mx	MSE My	MSE Mz	Train MSE	Val MSE	Notes
U-Net	0.202	0.125	0.121	0.359	~0.12	~0.21	Best overall; still improving at ep 30
FNO	0.258	0.125	0.121	0.528	0.087	0.305	Severe overfit; large train/val gap
Transformer	0.262	0.125	0.121	0.541	—	—	Best generalizer; val still dropping
Neural ODE	0.282	0.125	0.121	0.599	—	—	Slow; noisy validation curve
CNN (plain)	0.321	0.125	0.121	0.715	—	—	Effectively failed

* MSE Mx=0.125 and MSE My=0.121 are identical across ALL five architectures — no model learned the in-plane components. Only Mz shows architecture-dependent variation.

9. Key Findings & Failure Modes

#	Finding
1	RAAR phase retrieval failed as a seed (L2=100%, ang=90.9 deg). Phase ambiguity in 3D XMCD is not resolved by RAAR + shrink-wrap.
2	Direct LS (no phase retrieval) outperformed RAAR as a seed (55 deg vs 91 deg). Simpler approach was better.
3	RESIRE-V (Adam, 300 iters) reduced amplitude residual but angular/L2 error barely moved from seed (~47 deg final). Gradient descent stagnated.
4	Forward model self-consistency check FAILED (residual=0.342) — q-axis calibration mismatch between detector grid and volume grid in RESIRE-V.
5	Best 3D result: L2=83.5%, mean angular error=47.8 deg vs 90 deg random. Meaningful but not quantitative reconstruction.
6	ML benchmark: U-Net best (MSE=0.202) on 2D inverse scattering. No architecture recovered Mx or My — all collapsed to identical MSE ~0.125.
7	Mz most visible to sensing geometry (0.571 vs Mx=0.411, My=0.467). Insufficient angle diversity to disentangle in-plane components from amplitude alone.